

CHANGES IN THE CONCENTRATION OF MALATES, TARTRATES, AND TOTAL FREE ACIDS IN FLOWERS AND BERRIES OF *VITIS VINIFERA*¹

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Considerable data have been published on changes in the concentration of organic acids in grapes during ripening. However, relatively few studies have been reported, on such changes during the preripening period, and almost nothing has been reported about changes in concentration that occur during the periods of prebloom, bloom, berry set and shatter.

Several investigators have shown that concentrations of malic and tartaric acids and their acid salts decrease rapidly during maturation of the berry and that these changes vary greatly with variety and season (2, 15, 17). Berries of most varieties that have been investigated have their highest levels of malic and tartaric acids just prior to the véraison (4, 5, 8) i. e. just before the grapes change color and begin to accumulate sugar rapidly. Generally the amounts of malic and tartaric acids are not greatly different from each other during this period. During maturation of the fruit, however, malic acid generally decreases at a faster rate than does tartaric acid; so at the time of harvest tartaric acid is usually present in greater amount than malic acid (2, 4). Many varieties have been classified on the basis of low malic or high malic acid at maturity (1, 2, 4).

In grapes grown in cool regions, or in cool seasons, malic acid generally comprises a greater percentage of total acidity than it does in the same varieties grown in warmer regions or seasons. However, many exceptions to this relationship have been observed (2, 16). For a more thorough review of the literature pertaining to or-

ganic acid in grapes, see Genevois (9, 10), Peynaud (15), Amerine and Winkler (4, 5), and Amerine (1, 2).

The primary purposes of the present investigation were (1) to determine the period(s) in the growth cycle of grapevines during which malic and tartaric acids and their acid salts are synthesized in greatest amounts; (2) to ascertain during what period(s) they are broken down most rapidly; and (3) to compare differences among varieties in synthesis and breakdown of organic acids.

MATERIALS AND METHODS

Used in this investigation were eight varieties of *Vitis vinifera* L. in the University experimental vineyard at Davis, California (Table 1). To obtain flowers and berries of the same physiological age, buds were tagged on the day the first leaf began to unfold (budburst); only flowers and fruit from these shoots were sampled. Sampling of clusters began approximately 2 weeks after budburst. Two clusters per vine were taken from 20 vines of each of the eight varieties approximately every 7 to 11 days. The samples were placed in polyethylene bags, sealed, immediately frozen with powdered dry ice, and stored at -20°C until analyzed. When the berries reached a weight of 0.3 to 0.4 g, they were sampled on an individual basis rather than on a cluster basis. Two to four berries per cluster from five clusters on each of 20 vines per variety were sampled. Samples were taken randomly from all parts of the vine and cluster.

Fifty grams of flowers or berries were extracted with ethanol by a procedure described previously (11). The flower was considered to include the calyx, calyptra, stamen, pistil, and pedicel. The stamens and pistil had not yet fully developed in

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TABLE I
Seasonal Changes in Total Acidity of Flowers and Berries from Eight Grape Varieties

Date sample taken (1964 season)	Stage of flower or berry development	Part of cluster analyzed	Variety							
			Thompson Seedless	White Riesling	Emerald Riesling	Semillon	Sylvaner	French Colombard	Barbera	Carignane
			Total acid (g tartaric/100 g fresh tissue)							
April 20	Pre-bloom	Flower	0.85	0.94	0.95	0.98		0.80	0.88	
April 29	Pre-bloom	Flower	1.58	1.54	1.49	1.55	0.74	1.62	1.35	0.78
May 11	Pre-bloom	Flower	2.47	1.83	1.90	1.94	1.75	1.57	2.02	2.07
May 20	Bloom	Flower	2.39	2.67	2.81	2.63	2.44	2.56	2.70	2.56
May 29	Bloom & berry set & shatter	Flower & or	2.20	2.15	2.40	2.25	2.20	2.35	2.20	2.15
June 10	Shatter & green berry	berry	1.70	1.74	1.78	1.65	1.70	1.78	1.78	1.75
June 17	Green berry	Flower & or	2.25	2.07	2.06	2.26	2.17	2.06	2.42	1.93
June 24	Green berry	berry	2.78	2.26	2.11	2.38	2.17	2.30	2.50	2.26
July 1	Green berry	Berry	3.04	2.54	2.19	2.46	2.18	2.41	2.74	2.48
July 8	Green berry	Berry	3.07	2.54	2.30	2.65	2.42	2.52	2.84	2.75
July 15	Green berry	Berry	3.62	2.94	2.85	2.68	2.63	2.82	2.90	2.88
July 22	Green berry & véraison	Berry	3.22	3.03	2.84	2.80	2.61	2.72	3.11	3.08
July 29	Green berry & véraison	Berry	1.94	2.58	3.48	2.40	2.07	2.72	3.11	3.07
Aug. 5	Green berry & véraison & ripening	Berry	1.12	2.49	2.73	1.48	1.69	1.98	2.99	3.38
Aug. 12	Véraison & ripening	Berry	0.85	1.52	2.26	1.14	1.55	1.68	2.40	2.34
Aug. 19	Véraison & ripening	Berry	0.62	1.13	1.78	0.78	0.84	1.18	1.68	1.96
Aug. 27	Ripening	Berry	0.58	0.84	1.11	0.60	0.77	1.04	1.16	1.47
Sept. 5	Ripening & ripe	Berry	0.58	0.73	0.90	0.56	0.65	0.93	1.12	1.21
Sept. 12	Ripening & ripe	Berry	0.52	0.62	0.80	0.45	0.68	0.92	1.04	1.23
Sept. 18	Ripening & ripe	Berry	0.50	0.53	0.63	0.48	0.65	0.93	1.04	1.19
Sept. 25	Ripe	Berry	0.37	0.52	0.58	0.38	0.53	0.68	0.76	0.77
Oct. 2	Ripe & over ripe	Berry	0.37	0.49	0.58	0.37	0.50	0.72	0.64	0.67
Oct. 9	Ripe & overripe	Berry	0.41	0.48	0.59	0.33	0.50	0.61	0.63	0.62
Oct. 20	Overripe	Berry	0.31	0.39	0.49		0.41	0.55		0.52

clusters sampled April 20 and 29. The flowers were separated readily from the rachis by shaking while still frozen. The final concentration of the plant extract was 50 g (fresh weight) per 500 ml of aqueous ethanol solution (20 to 25% ethanol).

Total acids were determined on 10- or 20-ml samples of the ethanol extracts diluted to 200 ml with freshly boiled distilled water and titrated with 0.05N NaOH, using phenolphthalein as the indicator. The results are expressed as grams of tartaric acid per 100 g fresh weight of tissue. The total titratable acidity data (Table 1), expressed as a per cent of the fresh weight of the berries, are slightly less than if the data were expressed as g/100 ml juice, because of the relatively low acidity of the seeds and skin.

Malate¹ was determined by the highly specific malic dehydrogenase method of Mayer and Busch (12), using a model PM QII Zeiss spectrophotometer for detecting change in absorbancy. This method can detect as little as 15 ppm of malate, and is about 1000 times as sensitive as previously used methods (14, 18) for determining malate in grapes and wine.

To use the enzymatic method for determining malic acid in flowers and young berries, it was necessary to remove the pigments and other unidentified substance(s). This was done as follows: A sample (2 to 4 ml) of the extract, diluted about 5-fold with water, was boiled with approximately 0.2 g of charcoal (Darco G-60) for 3 minutes on a hot plate. The charcoal had been washed previously with 0.1% solution of malic acid, as well as washed exhaustively with distilled water. The charcoal was removed by filtration, with the aid of a vacuum, through Hirsch funnels containing 2.4-cm Whatman GF/A glass filter paper. After the charcoal was washed with about 50 ml of distilled water, the filtrate was made up to 100 ml. Aliquots (1 or 2 ml)

of the final diluted extracts were used for determination of malate.

Extracts of berry samples taken after July 1 did not inhibit the malic dehydrogenase reaction. Therefore, the charcoal treatment was not used after this date. Standard malic acid curves were prepared with and without going through the preliminary charcoal treatment.

Tartrate was determined by the vandate method (3), modified by the use of charcoal (Darco G-60) which had been washed with 0.1% tartaric acid and rinsed free of any unbound acid with distilled water. Five-to-10-ml samples of the alcoholic extracts were boiled 3 to 5 minutes in the presence of about 0.3 g charcoal, and then filtered through Hirsch funnels containing 2.4-cm Whatman GF/A glass filter paper. These procedures avoided loss of tartrates which was observed in preliminary experiments with unwashed charcoal and Whatman No. 2 filter paper. It should be noted that the vandate method of determining tartrate includes free tartaric acid as well as any salts of tartaric acid that may be present. All tartrate and malate data are the average of duplicate determinations.

In the present report and the one following, green, ripening, ripe, and overripe are used to describe stages of berry growth. These terms have been defined by Winkler (20). To avoid ambiguity, these terms as used in this report are defined on the basis of degree Balling as follows: green, 3 to 6.9°B; ripening, 7.0 to 19.9°B; ripe, 20.0 to 23.7°B; overripe, 23.8°B or greater. All varieties sampled Oct. 20th, regardless of °B, were overripenes. White Riesling was the only variety that did not fit into this classification, since the fruit was obviously overmature on Oct. 2, when the degree Balling was 22.5 (See table 5 in following report for reference to date and degree Balling).

RESULTS

The data in table 1 show the changes in total acidity over the entire growing season (April 20 to October 20). It should be noted that there is considerable overlap-

¹ The forms malate and tartrate in this communication include malic and tartaric acids as well as their respective salts.

DISCUSSION

ping of the varieties as to the date and indicated stage of flower or berry development.

The average dates of budburst were as follows: Thompson Seedless, French Colombard, and Barbera, March 30; White Riesling, Emerald Riesling, Semillon, and Sylvaner, April 3; Carignane, April 6. The bloom period for the eight varieties as a group began May 16 and extended to June 5. The average period of bloom for any one variety was about 10 days. Tables 2 and 3 indicate a sharp increase in tartrate and malate at initial phase of bloom period. This was followed by a decline toward the end of the bloom period and continued through berry set and shatter. The change in tartrate during this period was especially remarkable, increasing 200 to 600% during a 2-week period. The concentration of tartrate at this time ranged between 3 and 5 times that of malate (Table 4). Following berry shatter there was again a sharp rise in both these substances, reaching a peak about the middle of July. From near the end of June to the end of July, malates and tartrates were present in the green berries in about equal amounts (Table 4).

There was considerable variation among varieties as to the date that the berries began to change color. The dates were: July 22 to 29, Thompson Seedless; Aug. 5 to 12, White Riesling, Semillon, Sylvaner, French Colombard, Barbera, and Carignane; Aug. 19 to 27, Emerald Riesling. The *véraison* stage of berry development was accompanied by very rapid increases in sugars (table 5 in companion paper) and sharp reductions in tartrates, malates, and total acidity (Tables 1, 2, 3).

The change in malic acid concentration during ripening, ripe, and overripe periods of fruit development varied considerably among varieties. Those in the present study can be divided into two groups on the basis of the average tartrate/malate ratio at maturity (Table 4). Thompson Seedless, White Riesling, Emerald Riesling, and Semillon comprise a group of low-malate varieties, with an average ratio of 2.5 or greater and Sylvaner, French Colombard, Barbera, and Carignane are high-malate varieties, with a ratio of 1.5 or less.

Until about the end of July, total titratable acidity for all varieties (Table 1) compared quite closely to acidity found by adding values for tartrate and malate in tables 2 and 3. This indicates that most of the tartrate and malate in the berries up until the *véraison* was present as free tartaric and malic acids. However, after the berries began to change color there was a sharp drop in total titratable acidity, and in most varieties the sum of the tartrates and malates was considerably greater than the total acidity. These results may be explained by the extensive formation of the acid salts of tartaric acid and, to a much smaller extent, those of malic acid, during grape ripening. Bioletti *et al.* (8) found that as the Muscat variety approached maturity, the acid salts of tartaric acid, calculated as tartaric acid, approached total titratable acidity. This indicated the presence of very little free acid. Amerine (4) showed that only 10 to 20% of the tartrates were present as free acid at berry maturity in several *vinifera* varieties. A comparison of total acidity with the sum of the tartrates and malates indicates a marked difference in free acid among the varieties at berry maturity. Barbera and Carignane had relatively larger amounts of free acid, while Thompson Seedless and Semillon berries possessed considerably smaller amounts. Unfortunately, it was not possible to determine pH of the ethanol extracts accurately, and thereby calculate the amount of free malic and tartaric acids by the Henderson-Hasselbalch equation. Alcohol causes increased precipitation of acid tartrate and directly influences the activity of the hydrogen ions (4).

At maturity, the amount of malates present in the berries from the various varieties was a great deal more variable than the amount of tartrates. The concentration and activity of the enzymes involved in respiration of malic acid most likely differ among various varieties, and may account for the variations in per cent malic acid. Amerine (2) has suggested that a low level of malate in berries may be

TABLE 2
Seasonal Changes in Concentration of Tartrates in Flowers and Berries from Eight Grape Varieties

Date sample taken ^a (1964 season)	Variety							
	Thompson seedless	White Riesling	Emerald Riesling	Semillon	Sylvaner	French Colombard	Barbera	Carignane
	Tartrates as % tartaric acid (fresh weight basis)							
April 20	0.70	0.83	0.82	0.83		0.67	0.70	
April 29	0.99	1.01	1.03	1.06	0.40	0.93	1.00	0.33
May 11	1.93	1.42	1.54	1.40	1.40	1.18	1.63	1.75
May 20	1.67	2.37	2.19	1.90	1.88	1.90	2.29	1.90
May 29	1.61	1.61	1.98	1.50	1.55	1.85	1.58	1.43
June 10	1.31	1.51	1.50	1.25	1.50	1.15	1.50	1.45
June 17	1.30	1.33	1.35	1.17	1.45	1.20	1.29	1.23
June 24	1.35	1.02	1.02	1.33	1.10	1.30	1.08	0.99
July 1	1.37	1.10	0.98	1.29	1.06	1.34	1.24	1.19
July 8	1.34	1.25	1.75	1.45	1.62	1.68	1.62	1.42
July 15	1.78	1.50	1.73	1.27	1.52	1.80	1.50	1.40
July 22	1.56	1.45	1.60	1.37	1.29	1.60	1.52	1.45
July 29	1.20	1.52	1.66	1.29	1.10	1.67	1.61	1.47
Aug. 5	1.06	1.59	1.61	1.20	1.03	1.11	1.56	1.56
Aug. 12	0.95	1.28	1.50	1.10	1.06	1.10	1.28	1.39
Aug. 19	0.60	0.69	0.79	0.51	0.55	0.58	0.66	0.69
Aug. 27	0.58	0.58	0.67	0.45	0.58	0.60	0.58	0.61
Sept. 5	0.55	0.57	0.68	0.50	0.68	0.61	0.70	0.60
Sept. 12	0.58	0.51	0.61	0.54	0.63	0.51	0.60	0.62
Sept. 18	0.60	0.50	0.65	0.58	0.58	0.59	0.65	0.65
Sept. 25	0.62	0.57	0.69	0.55	0.62	0.61	0.69	0.71
Oct. 2	0.60	0.63	0.64	0.56	0.66	0.62	0.63	0.69
Oct. 9	0.65	0.61	0.65	0.55	0.62	0.60	0.68	0.67
Oct. 20	0.55	0.52	0.66		0.65	0.56		0.66

^a The stage of berry development and the part of the cluster analyzed for each date are given in table 1.

TABLE 3
Seasonal Changes in Concentration of Malates in Flowers and Berries from Eight Grape Varieties

Date sample taken ^a (1964 season)	Variety							
	Thompson Seedless	White Riesling	Emerald Riesling	Semillon	Sylvaner	French Colombard	Barbera	Carignane
	Malates as % malic acid (fresh weight basis)							
April 20	0.22	0.21	0.24	0.25	0.14	0.15	0.22	0.18
April 29	0.51	0.43	0.42	0.47	0.30	0.40	0.28	0.30
May 11	0.64	0.43	0.41	0.45	0.38	0.39	0.32	0.36
May 20	0.62	0.48	0.65	0.69	0.47	0.74	0.65	0.61
May 29	0.54	0.35	0.56	0.60	0.35	0.36	0.36	0.70
June 10	0.35	0.47	0.49	0.57	0.48	0.57	0.46	0.45
June 17	0.63	0.58	0.35	0.67	0.72	0.63	0.80	0.60
June 24	1.10	1.14	0.80	0.90	0.84	0.86	1.02	0.85
July 1	1.62	1.32	0.70	1.18	0.96	0.88	1.04	1.00
July 8	1.63	1.35	0.85	1.52	1.30	1.31	1.33	1.32
July 15	1.90	1.32	0.97	1.28	1.30	1.18	1.45	1.60
July 22	1.80	1.26	1.17	1.15	1.20	1.21	1.42	1.65
July 29	1.28	0.96	1.83	0.87	1.16	0.98	1.42	1.68
Aug. 5	0.60	0.94	1.08	0.76	0.81	1.01	1.36	1.70
Aug. 12	0.54	0.79	0.98	0.66	0.85	0.97	1.30	1.59
Aug. 19	0.41	0.54	1.00	0.68	0.69	0.81	1.09	1.35
Aug. 27	0.23	0.42	0.91	0.41	0.67	0.71	0.68	0.94
Sept. 5	0.27	0.29	0.47	0.32	0.58	0.62	0.67	0.84
Sept. 12	0.30	0.20	0.33	0.35	0.61	0.70	0.69	0.74
Sept. 18	0.19	0.18	0.27	0.23	0.55	0.58	0.42	0.72
Sept. 25	0.20	0.25	0.28	0.25	0.50	0.58	0.44	0.62
Oct. 2	0.22	0.22	0.19	0.24	0.33	0.61	0.46	0.51
Oct. 9	0.20	0.19	0.21	0.15	0.37	0.54	0.45	0.47
Oct. 20	0.12	0.06	0.08		0.36	0.50		0.38

^a The stage of berry development and the part of the cluster analyzed for each date are given in table 1.

TABLE 4
Seasonal Changes in Tartrate/Malate Ratio of Flowers and Berries from Eight Grape Varieties

Date sample taken ^a (1964 season)	Variety							
	Thompson Seedless	White Riesling	Emerald Riesling	Semillon	Sylvaner	French Colombard	Barbera	Carignane
	Tartrate/malate ratio							
April 20	3.18	3.95	3.42	3.32		4.46	3.18	
April 29	1.94	2.34	2.45	3.40	1.34	2.32	3.57	1.10
May 11	3.01	3.30	3.75	3.11	3.68	3.02	5.09	4.86
May 20	2.69	4.93	3.36	2.75	4.00	2.56	3.52	3.11
May 29	2.98	4.60	3.53	2.50	4.42	5.13	4.38	2.04
June 10	3.75	3.21	3.06	2.19	3.12	2.01	3.26	3.22
June 17	2.06	2.29	3.85	1.74	2.01	1.90	1.61	2.05
June 24	1.22	0.89	1.27	1.47	1.30	1.51	1.06	1.16
July 1	0.84	0.83	1.40	1.09	1.10	1.52	1.19	1.19
July 8	0.82	0.93	2.05	0.95	1.25	1.28	1.22	1.23
July 15	0.94	1.14	1.78	0.99	1.17	1.52	1.03	0.88
July 22	0.87	1.15	1.37	1.19	1.08	1.32	1.07	0.88
July 29	0.94	1.58	0.91	1.48	0.95	1.70	1.13	0.88
Aug. 5	1.76	1.69	1.49	1.57	1.27	1.10	1.15	0.92
Aug. 12	1.76	1.62	1.53	1.66	1.24	1.13	0.98	0.87
Aug. 19	1.46	1.28	0.79	0.75	0.80	0.72	0.61	0.51
Aug. 27	2.52	1.38	0.74	1.09	0.86	0.84	0.85	0.64
Sept. 5	2.04	1.96	1.44	1.56	1.17	0.98	1.04	0.71
Sept. 12	1.93	2.55	1.85	1.54	1.03	0.73	0.87	0.84
Sept. 18	3.16	2.78	2.40	2.52	1.05	1.02	1.55	0.90
Sept. 25	3.10	2.28	2.46	2.20	1.24	1.05	1.57	1.14
Oct. 2	2.73	2.86	3.37	2.33	2.00	1.02	1.37	1.35
Oct. 9	3.25	3.21	3.10	3.66	1.67	1.11	1.38	1.42
Oct. 20	4.58	8.66	8.25		1.80	1.12		1.73

^a The stage of berry development and the part of the cluster analyzed for each date are given in table 1.

due to decreased translocation of malate into the fruit from the leaves.

A rapid decrease in concentration of malate and tartrate during berry ripening was noted in all varieties investigated (Tables 2 and 3). The decrease in these acids is generally attributed to: (a) increased respiratory rate, (b) reduction in the amount of acid translocated from the leaves, (c) dilution effect due to increased volume of the fruit, and (d) increase in level of potassium in fruit with maturity. The reduction of malic acid in the berry as it ripens could also be partly due to a loss in activity of the CO_2 -fixing mechanism or to the loss of enzymic activity necessary for reactions intermediate to malate synthesis. It also might be possible that malic and other organic acids are used for synthesis of carbohydrates, as has been shown by Benedict and Beevers (6) for castor bean.

The biochemical pathways for the formation of malic and tartaric acids in grapes are not known. The fact that these acids accumulate to such high concentrations indicates participation of reaction other than those in the tricarboxylic acid cycle. Malic enzyme, which catalyzes the synthesis of malic acid by the beta-carboxylation of pyruvic acid, has been shown to be present in grape berries (19). Phosphoenolpyruvic carboxylase and phosphoenolpyruvate carboxykinase also are able to form malic acid via oxalacetic acid. More recently, malate synthetase, which catalyzes the condensation of acetyl CoA and glyoxalate to form malate, has been found in plants (20). Payes and Laties (13) have shown another way that malic acid can be synthesized in plants from gamma-hydroxy-alpha-ketoglutarate. The enzymic formation of the latter compound from glyxalate and pyruvate has recently been demonstrated in plant extracts (13). The enzymes and intermediate precursors for synthesis of D-tartaric acid in plants are not known.

Several investigators (2, 15, 17) have shown that the level of malates and tartrates for a particular grape variety are influenced considerably by season and region. Other factors, such as level of crop and diseases, also probably influence the ratio of these acids. For these reasons,

classification of grape varieties based on tartrate/malate ratio should be done with caution.

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SUMMARY

In eight grape varieties studied from April 20 to October 20, tartrate, malate, and total titratable acids increased rapidly in the flowering parts of the vine up to early bloom. The levels declined during late bloom and berry set. During prebloom and bloom periods, tartrates were present in amounts two to five times as great as the amounts of malate. After shatter, concentrations of both acids in the berries increased rapidly for about 4 weeks, a period corresponding closely to the period of rapid growth of berries.

Tartrates and malates remained at a high level in about equal amounts until the berries began to change color and ripen. The ripening period (July 22 to Sept. 18) was associated with sharp decreases in malates and tartrates in all varieties. During the ripe and overripe stage of maturity, tartrates remained at about the same level, while malates and total titratable acidity continued to decrease.

The varieties studied were classified into two groups according to the average ratio of tartrate to malate (T/M) at berry maturity. Low-malate varieties (T/M ratio greater than 2.5) included Thompson Seedless, White Riesling, Emerald Riesling, and Semillon. High-malate varieties (T/M ratio less than 1.5) included Sylvaner, French Colombard, Barbera, and Carignane.

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